DDM Exam — Part 5 Answers

Single Choice & True/False Questions (20 Points)

Section 1: Single Choice Questions (10 Points)

Q	Statement	Correct Answer	Why
a	Multiple processors may access the same DBMS buffer (main memory) in the...	**✓ Shared-Everything-Architecture**	In Shared-Everything, all CPUs share a single main memory / DBMS buffer. Shared-Nothing has separate memory per node; Shared-Disk shares disks but not RAM.
b	The fact that redundant copies of tables/fragments are visible to the user as only one table/fragment is referred to as...	**✓ Replication Transparency**	Replication Transparency hides the existence of multiple copies from the user. Fault-Tolerance is a benefit, not a transparency type. One-Copy-Serializability is a consistency property.
c	Horizontal fragmentation decomposes a relation...	**✓ by using the selection operation σ into sets of tuples**	Horizontal fragmentation uses σ (selection) to split rows into subsets. Projection π is used for vertical fragmentation. Intersection ∩ is not a fragmentation operator.
d	The usage of several fragmentation methods on one relation is called...	**✓ Hybrid Fragmentation**	Hybrid Fragmentation combines horizontal and vertical fragmentation on one relation. Derived Fragmentation is horizontal fragmentation based on another relation's partitioning.
e	During query optimization, using a description of the content of a relation by a logical condition is referred to as...	**✓ Qualified Relations**	Qualified Relations annotate fragments with logical conditions (qualifications) that describe their content, enabling elimination of unnecessary fragment accesses during optimization.
f	Given R(A,B,C), S(C,D,E) and T(E,F,G), an undesirable join order containing a Cartesian Product is...	**✓ R ■ T ■ S**	R and T share no common attributes (R has C, T has E — no overlap), so R ■ T produces a Cartesian Product. R■S (on C) and T■S (on E) are valid joins.
g	The Selectivity Factor of an equivalence condition A = v can be estimated as...	**✓ SF(A=v) = 1 / val(A)**	val(A) is the number of distinct values of A. Assuming uniform distribution, each value has probability 1/val(A). The other formulas are for range conditions (A > v or A < v).
h	Full One-Copy-Serializability is granted for replicated data by...	**✓ the Read One Write All-approach**	ROWA executes all writes within a single transaction context across all copies, guaranteeing 1-Copy-Serializability. Primary Copy uses asynchronous propagation so secondary copies may lag (snapshot semantics).

Q	Statement	Correct Answer	Why
i	The term 'Snapshot Semantics' in Primary Copy-replication refers to the fact that...	**✓ secondary copies may be out of date with the primary copy**	In Primary Copy, updates propagate asynchronously to secondaries. Reads from secondaries reflect a consistent but possibly older state — a 'snapshot' of a past point in time.
j	To compute a join result, the Ship Whole-strategy...	**✓ always transfers one relation in its entirety**	Ship Whole moves one complete relation to the other node. Semi-Join transfers only the join column first. Bit-Vector Join transfers a compact bit vector (hash filter).

Section 2: True or False Questions (10 Points)

#	Statement	Answer	Explanation
1	Federated Database Systems are a solution to integrate previously existing and possibly distributed databases.	**TRUE**	Federated DBS are specifically designed to integrate pre-existing, heterogeneous, and possibly distributed databases while preserving their autonomy.
2	The reconstruction operation for fragments created by vertical fragmentation is the Semi-Join.	**FALSE**	Vertical fragments are reconstructed using a Natural JOIN (■) on the common key attribute, not Semi-Join. Semi-Join is a join strategy for distributed query processing.
3	An optimal horizontal fragmentation can be derived from the analysis of typical queries and their frequencies.	**TRUE**	Horizontal fragmentation is designed based on predicate usage analysis — examining which conditions appear in queries and how often, to create fragments that minimize cross-node data access.
4	A disadvantage of the Bit Vector Join is that tuples not required for a join are possibly transferred over the network.	**FALSE**	That disadvantage belongs to Ship Whole. The Bit Vector Join's disadvantage is false positives — the hash function may match tuples that are not actual join partners (hash collisions).
5	The Ship Whole-strategy is always preferable to the Fetch as Needed-strategy.	**FALSE**	Neither strategy is always better. Ship Whole is worse when most tuples from the shipped relation are not needed. Fetch as Needed is worse when many individual lookups are needed (high message overhead).
6	The Fetch as Needed-strategy is always preferable to the Ship Whole-strategy.	**FALSE**	Same reasoning as above — no strategy is universally superior. The best choice depends on join selectivity and message overhead relative to data transfer size.
7	Deadlocks are problematic in a distributed database, because according wait relationships cannot always be detected locally.	**TRUE**	In distributed systems, a deadlock cycle can span multiple nodes. Each node only sees its local wait graph, so a global cycle may be invisible locally — making detection significantly harder.
8	For Timestamp-based synchronization transactions are assigned a unique timestamp at the beginning.	**TRUE**	Each transaction receives a unique timestamp at its start (Begin of Transaction). This timestamp is used to order conflicting operations and decide which transaction must abort.

#	Statement	Answer	Explanation
9	The 2-Phase-Commit Protocol grants Availability for distributed transactions.	**FALSE**	2PC guarantees Atomicity and Durability (all-or-nothing commit), NOT Availability. In fact, 2PC can block indefinitely if the coordinator fails, reducing availability.
10	For Cost-based Optimization less possible plans have to be considered in Distributed Databases, e.g. because Replication leads to less redundant plans.	**FALSE**	It is the opposite — replication increases the number of plans to consider, since data may exist on multiple nodes and each copy is a valid access option, multiplying the plan space.

Source: DDM Lecture Slides, Prof. Dr. Eike Schallehn, OvGU Magdeburg